

The Cost of Not Being Seen:

Automated Licence-Plate Reader Exposure on American Commutes, and the Rising Price of Avoiding It

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Abstract

Automated licence-plate readers (ALPR) have spread from tollbooths to residential streets; one vendor, Flock Safety, holds 80 % of the 142,991 installations mapped in North America. We measure what that means for an ordinary trip, and how fast it is changing. A continental routing engine models each camera’s field of view and can find a route that crosses none. We route 56,131 home–work commutes from all fifty states and the District of Columbia, with and without avoidance. 75.3 % of commuters pass at least one reader, a median of three. Refusal is cheap: 84.2 % can reach work past none, and avoiding adds a median of 1.62 min (3.14 min for commuters with a camera to avoid). OpenStreetMap keeps every edit, so we rebuild the camera map as it stood on the first of every month since January 2024—exactly, node for node, from the full edit history—and re-ask the question of each: the same commutes on the same roads, re-scored against every monthly map and re-routed against every quarterly one. The map grew 128-fold in thirty-three months, and measured exposure almost in proportion, from 2.2 % of commuters passing a mapped reader to 75.3 %. Refusal grew dearer per camera as well as in total: on the sparse maps of 2024 95 % of exposed commuters could reach zero, for a median of 0.8 min; today 79 % can, for 3.1 min, and the price of evading each camera has nearly doubled. What a commute passes, and what declining costs, are functions of a map that is still filling in; we say when ours was drawn.

Keywords: automated licence-plate recognition; surveillance; route choice; volunteered geographic information; OpenStreetMap; edit history.

1 Introduction

Automated licence-plate readers (ALPR) photograph every passing vehicle, resolve the plate, and retain the read [14]. Unlike a traffic camera they are not triggered by an offence, and unlike a patrol officer they do not tire. Fixed, pole-mounted units have spread from limited-access highways into residential neighbourhoods, arterial roads, and the entrances of private subdivisions [27, 39].

The public debate about this expansion is conducted in the vocabulary of policy: retention windows, warrant requirements, audit logs, data-sharing agreements. What that vocabulary lacks is a measurement of the thing being regulated. How much of this does a person actually drive through on an ordinary Tuesday? And if they would rather not, is declining a practical option or a theoretical one?

This paper answers both, and does so by building the instrument the question implies. We construct a routing engine that treats ALPR fields of view as first-class geometry and can be asked

for a route that avoids them. Running each sampled commute through it twice—once normally, once avoiding—yields the exposure and the price of refusing it, for the same trip, on the same network, at the same speeds.

The second quantity is where the paper earns its keep. A privacy protection that costs four minutes is a feature. One that costs forty is available only to people with forty minutes to spare. We therefore report the price of refusal as a distribution rather than a point: for most commuters it is a few minutes, and for a tail it is not.

The third question follows from the data source, and it is the one this paper is about. The cameras are mapped by volunteers, and the map has grown 128-fold since January 2024. OpenStreetMap keeps every edit, so the measurement can be repeated on the map as it stood at any past instant. We repeat it on the first of every month for thirty-three months: the same commutes on the same roads, re-scored against each month’s cameras and re-routed against each quarter’s. How much surveillance a commuter drives through, and what declining it costs, turn out to be functions of how much of it had been mapped—and both have risen on every map since.

1.1 Contributions

- **A directional exposure model at continental scale.** Cameras are wedges, not circles. 98.0 % of mapped North American ALPR installations carry a bearing tag, making this feasible nationally. Proximity-based counting inflates exposure by charging drivers for cameras aimed at the opposite carriageway; we do not do it.
- **Avoidance routing with no forked router.** 157,084 camera cones, unioned into 135,210 polygons, are compiled into a stock GraphHopper spatial index at import time, so an ordinary per-request custom model can reference them by name. No engine patches, no geometry in the request.
- **A commute-weighted rather than area-weighted sample.** 56,131 real home–work pairs in all fifty states and the District of Columbia, drawn from LEHD LODES with probability proportional to worker count and weighted to each state’s commuters.
- **A vendor-resolved account of exposure.** Flock’s cameras and other makers’ are bought by the same kind of customer—police departments, where the operator is known—but stand differently: other makers’ units are more often multi-headed and stand beside the busiest roads, and per unit deployed they generate 2.26 times as many camera encounters.
- **The camera map at any instant, from the edit history.** Every version of every node ever tagged as a reader, extracted once from OpenStreetMap’s full-history dump and validated node for node against the public history server, gives the map on the first of every month since January 2024 and the day each camera entered it. We quantify the 7.1-fold per-capita spread in mapped density between states, in which mapping effort and deployment cannot be told apart, and check the map’s coverage against the counts 1,050 police departments publish, which it holds a median 98 % of.
- **A thirty-three-month trend on a fixed sample.** The same commutes on the same roads against thirty-three monthly maps, re-scored for exposure on each and re-routed for avoidance on each quarter’s: exposure rose almost in proportion to the map, and refusal grew dearer per camera as well as in total.

2 Related work

This paper sits where three bodies of work meet: research on ALPR as a policing technology, on volunteered geographic data and its biases, and a small practice of routing around surveillance. We take them in turn, and then state what is, to our knowledge, new here.

ALPR as a policing technology. Criminological research on ALPR has mostly asked whether it works. Lum et al. [26] tested ALPR-equipped patrols in crime hot spots with a place-based randomized design across two jurisdictions, and Lum et al. [28] review the impact studies since, asking whether the technology’s cost-effectiveness can be established at all. Shjarback and Sarkos [38] evaluate the Atlantic City Police Department’s installation of fixed readers on all entrances and exits to the city, reporting reductions in shootings, motor-vehicle theft and property crime but not in violent crime overall. Sociological and legal work asks what the technology does to policing itself. Brayne [2]’s fieldwork in the Los Angeles Police Department shows plate readers lowering the threshold for inclusion in police data to people with no police contact, and Joh [24] and Ferguson [12] describe the automated suspicion and data-driven policing such systems enable. Monahan [30] examines Flock’s network as a communicative infrastructure through which discriminatory policing crosses jurisdictions. The legal status of the resulting location records is unsettled: in *Carpenter v. United States* the Supreme Court held that obtaining historical cell-site location records is a Fourth Amendment search [3], and Finklea [14] reviews the questions law-enforcement ALPR use raises. None of this work measures how much ALPR surveillance an ordinary trip passes.

Volunteered geographic data. The camera map is volunteered geographic information [19], whose quality has been assessed against authoritative data since OpenStreetMap’s early years [21] and whose completeness is known to vary with place and development [22]. For surveillance specifically, the Atlas of Surveillance records which agencies use which technologies but not where devices stand [9]; DeFlock supplies the locations [4]. Coverage bias is the central threat to every result built on such data, and we know of no earlier measurement of it for ALPR; we compare the map with the camera counts police departments publish in Flock’s transparency portals (Section 8.2), and its own history shows how far it was from complete on any earlier date.

Exposure along the way. That exposure to a place-bound feature depends on where people go and not only on where they live is the uncertain geographic context problem [25]; measuring surveillance along the commute rather than at the home address is an answer to it, and the unit of analysis throughout is the trip.

Routing around surveillance. A route that avoids cameras is an older idea than ALPR: in 2001 iSee computed “paths of least surveillance” around Manhattan’s closed-circuit cameras [23, 37], of which a civil-liberties survey had counted some 2,400 visible from the street [31]. Several services now route drivers around mapped ALPR, some using each camera’s field of view [1, 5, 6, 15], and in network measurement Edmundson et al. [7] characterize, and try to avoid, Internet paths through surveillance states. None publishes a measurement of how much surveillance ordinary trips pass or what avoiding it costs; the routing engine here is the instrument for that measurement, not its contribution.

What is new here. To our knowledge this paper is the first to measure, nationally and on real commute flows, how many ALPR fields of view commuters pass and what avoiding them costs; the

first to reconstruct the crowdsourced ALPR map from OpenStreetMap’s edit record at monthly resolution and follow a fixed sample of trips across it; and the first to check that map’s coverage against the camera counts police departments publish.

3 System

3.1 Why a fork, and of what

The measurement needs a router that treats “avoid these 135,210 polygons” as a cost rather than a post-hoc filter, and the eventual application needs to be a real navigation client rather than a research script.

OsmAnd [36] is the base: a mature open-source Android navigation client with first-class OpenStreetMap support, offline maps, and a documented online-routing extension point. Its *code* is GPLv3; its *artwork* is CC-BY-NC-ND 4.0 and must therefore be replaced rather than modified in a derivative.

GraphHopper 11.0 [20] is the routing engine. Valhalla was evaluated and rejected: its `exclude_polygons` support has a record of routes passing *through* the excluded region, one a genuine defect fixed only in December 2025 [43] and another report still open [42]—and a route through an excluded region is precisely the error everything here depends on not making.

One architectural property made the rest tractable. OsmAnd’s online routing engines return *geometry only*; turn instructions are re-derived client-side from offline map data. A custom online router therefore never has to synthesise navigation instructions.

3.2 The keystone: import-time custom areas

GraphHopper resolves `custom_areas.directory` into a spatial index *at graph import time*. We verified empirically—the documentation is ambiguous—that an area registered this way can be referenced by name from a *per-request* custom model:

```
{ "priority": [ { "if": "in_alpr", "multiply_by": 0.01 } ] }
```

This is what makes the design possible without forking the engine; shipping 135,210 polygons in every routing request is not viable. The cost is that regenerating cones requires a full re-import (82min to 113min for North America, by load).

Two further configuration decisions are load-bearing:

- **Contraction Hierarchies are disabled.** CH [16] bakes edge weights into the prepared graph, which is incompatible with a per-request avoidance strength. Landmarks (LM, the ALT algorithm of Goldberg and Harrelson [18]) are used instead.
- **The `car_alpr` profile shares car’s LM preparation.** This halves preparation time and is valid *only* because avoidance multiplies priority strictly downward, so the borrowing profile’s edge weights never fall below the preparation profile’s: the landmark distances remain lower bounds, the condition LM correctness requires.

3.3 Deployment posture

The routing server binds to localhost. It has no authentication and no rate limiting and is not internet-facing. Any public deployment must sit behind a reverse proxy supplying both.

Table 1: Directional and vendor tagging across the two 2026 snapshots. The direction-key rows overlap (392 and 415 nodes carry both keys). Vendors are matched as case-insensitive substrings of the first of the four vendor keys present, in both columns.

Property	2026-07-24		2026-09-22	
	Count	Share	Count	Share
Camera nodes	120,838		142,991	
Any direction tag	119,249	98.7 %	140,112	98.0 %
via <code>direction</code>	117,356	97.1 %	138,216	96.7 %
via <code>camera:direction</code>	2,285	1.9 %	2,311	1.6 %
Multi-head (;-separated)	7,014	5.8 %	9,150	6.4 %
Arc-range form (199–269)	4,806 tokens	—	4,632 tokens	—
<code>operator</code> tagged	22,354	18.5 %	23,872	16.7 %
Flock Safety	99,650	82.5 %	114,360	80.0 %
Motorola Solutions	5,993	5.0 %	7,411	5.2 %
Genetec	2,876	2.4 %	3,721	2.6 %
Leonardo	1,020	0.8 %	1,215	0.8 %
Axis Communications	1,052	0.9 %	3,327	2.3 %
No vendor tag	5,811	4.8 %	6,850	4.8 %

4 Data

All sources are public and key-free.

4.1 ALPR installations

Source: OpenStreetMap [33], queried through the Overpass API [32] for nodes tagged `man_made=surveillance` with `surveillance:type=ALPR`, matched case-insensitively (19 readers are tagged `alpr`)—the corpus surfaced by the DeFlock project, taken from the upstream rather than a mirror. Retrieved in five regional queries (continental US, Alaska, Hawaii, Canada, Mexico) and deduplicated by OSM node id. The results in this paper rest on the snapshot of **2026-09-22: 142,991 unique camera nodes**, 142,257 of them in the 50 states and the District of Columbia (Figure 1); the rest fall in the parts of Canada and Mexico the regional queries cover. A first snapshot, taken 2026-07-24, held 120,838 (120,272 in the states), and Section 4.3 reconstructs the map on the first of every month since January 2024.

A fetch that fails over between Overpass mirrors has to check the mirror’s database date. On the first attempt at the September snapshot one regional query fell through to a mirror whose database was dated 2026-05-06; that box overlaps the southern United States, regions were merged last-wins, and roughly 13,600 cameras across Texas, Arizona and Florida silently reverted to four-month-old tags while some 400 nodes deleted since May returned as ghosts. Nothing failed. The fetch now rejects any answer older than 48 hours and merges overlapping regions newest-database-first; every tile of the snapshot used here falls within a 14-minute window.

Three tagging realities shaped the parser, each contradicting a plausible assumption:

1. **direction, not camera:direction.** The OSM wiki documents `camera:direction` for surveillance nodes [35]. Mappers overwhelmingly use plain `direction`, by a factor of 60. Reading only the documented key discards 98 % of the bearings.
2. **Arc ranges are real, and mostly 45°.** 4,632 values, on 4,372 nodes, give an explicit range;

(a) 142,257 mapped ALPR installations, by vendor

(b) Installation density (log scale)

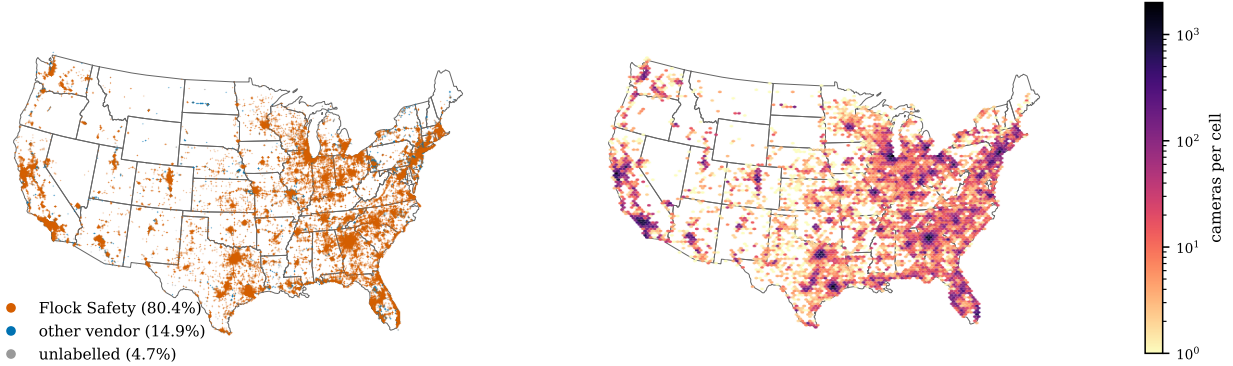


Figure 1: Mapped ALPR installations in the contiguous United States, 22 September 2026. (a) By vendor: Flock Safety dominates by count almost everywhere. (b) Density on a log scale. The visible structure is partly deployment and partly *mapping*: see Sections 6.3 and 8.2. Camera data © OpenStreetMap contributors.

Table 2: Vendor-split cone sets, 2026-09-22 snapshot; cameras are those with a bearing the parser can read (140,043 of 142,991). The cones-per-camera column is itself a result: non-Flock installations are multi-head more than six times as often (22.8 % against 3.6 %). A Flock unit typically watches one approach; other vendors’ units more often watch several lanes or directions.

Set	Cameras	Cones	Cones/camera	Union parts
Flock Safety	113,611	120,366	1.06	109,427
Other named vendor	20,435	30,128	1.47	21,587
No vendor tag	5,997	6,590	1.10	5,211

3,806 of them (82 %) span exactly 45° . That is what justifies 45° as the default span for the $\approx 97\%$ of cameras giving a bare bearing—the default is inherited from the data, not assumed.

3. **Multi-head units exist** (320;190, 0;72;144;216;288, 199-269;325-35); 9,150 nodes. These typically watch *both* carriageways, so silently taking the first bearing is the worst available error: it makes a camera that sees everyone look like one that sees half. The parser emits one wedge per head.

4.2 Cone construction

Each camera head becomes a circular sector: apex at the node, bisector along the tagged bearing, span 45° (or the tagged arc), radius 60 m. The radius must reach across the carriageway from a set-back pole; it was chosen on that reasoning and then swept over $\{30, 45, 60, 90\}$ m (Section 8.3). Measured exposure is only mildly sensitive to it, but the avoidance result is not robust above the assumed value.

All wedges are unioned into a single MultiPolygon of **135,210 parts**, published as one feature with id `alpr`. Overlapping wedges merging is correct and intended: the question is *is this road segment observed*, not *by how many devices*.

Assigning vendor requires care. Vendor appears under four keys (`manufacturer`, `surveillance:`

`manufacturer`, `brand`, `surveillance:brand`) and is spelled inconsistently. An exact match on `manufacturer = "Flock Safety"` finds 109,892 cameras; a case-insensitive substring match on the first of the four keys present finds 114,360. The 4,468 difference is 3,866 nodes that give the vendor only under `brand`, 599 that spell the manufacturer differently (478 of them “Flock Group Inc.”), and 3 that use `surveillance:manufacturer`. Misfiling 3.9 % of Flock cameras as non-Flock is the one error a vendor comparison cannot absorb.

Untagged cameras are held out of *both* vendor sets rather than assigned to the complement of Flock: an untagged camera may well be an unlabelled Flock unit. They are reported as their own set so nothing is silently dropped, with the caveat that their direction coverage is lower (88.0 % against 99.4 % for Flock and 93.8 % for other named vendors—the last down from 98.5 % in July, as newly mapped Axis units often arrive without a bearing).

4.3 Camera maps of earlier dates

OpenStreetMap keeps every edit, so the camera map can be reconstructed as it stood at any past instant. The public Overpass server answers a query carrying a `[date:...]` qualifier from its history: nodes deleted since are restored, nodes added since are absent, and every tag has its value from that day. Asked for Georgia at the July snapshot’s timestamp it returned 9,742 nodes, the snapshot’s own count to the node. History queries for the whole continent time out on the public server, so a date is fetched state by state and merged by node identifier; DeFlock’s own Overpass mirror keeps no history and cannot serve them. They are heavy, and after one complete date the public server began refusing our connections.

Every other date therefore comes from OpenStreetMap’s full-history dump [34] (163 GB, 14 September 2026, checked against its published MD5), read once. Two passes collect every version of every node that was ever tagged as an ALPR—169,565 nodes and 248,769 versions, including the later versions that deleted or retagged them—and the map at any instant is then a filter over that file, with Overpass’s semantics: each node’s latest version at or before the instant, kept if it was visible and still tagged. On the one date both sources cover, 1 July 2025, they agree exactly—22,690 nodes, the same identifiers, the same tags, the same coordinates—and on 1 January 2026 the filter reproduces an earlier, independent pass over the dump node for node (63,526). The same file gives the day, to the second, on which each camera entered the map, and a monthly count of the map that needs no aggregating service.

We reconstruct the map on the first of every month from January 2024 to September 2026, thirty-three maps, and set them beside the study’s snapshots of 24 July and 22 September 2026. Cones are built from each exactly as from the current map (Section 4.2). The share of mapped cameras carrying a bearing, and so receiving a cone, was about 90 % in 2024 and 98 % in 2026: the earliest maps are slightly less complete even on their own terms. Exposure is measured on every monthly map; avoidance is routed on the first of every quarter (Section 5.4).

One caveat governs everything built on these maps. OpenStreetMap records when a camera was *mapped*, not when it was installed: an installation date is present on 47 of 142,991 nodes. A camera first mapped in 2026 may have been on its pole since 2021. A past map is therefore what a study run on that day would have seen, and the trend of Section 6.5 describes how the measured picture changed as volunteers filled in the map—most of it after DeFlock was founded in October 2024 [4]—not how fast cameras were deployed.

4.4 Road network

Geofabrik North America extract [17], 18 GB, retrieved 2026-07-24 and held fixed for every camera map, so that the trend of Section 6.5 isolates the camera data: every unavoided route is the same route on every date. Imported with `ignored_highways = footway,construction,cycleway,path,steps`. `track` is deliberately retained: unpaved tracks are plausible avoidance routes, and excluding them would bias the cost estimate downward by removing options a determined driver has.

4.5 Commute sample

Source: LEHD **LODES8** [41], file `od_main_JT00`: all jobs with both ends in-state, built from unemployment-insurance wage records linked to employer establishments. These are administrative counts of real jobs, not survey estimates. The year is 2022 for 49 of the 51 jurisdictions; LODES has not published 2022 for Michigan, which uses 2021, or for Alaska, whose latest year is 2016. LODES8 codes every year against 2020 census geography, so an older year still names current tracts.

Sampling design carries more weight here than sample size. Uniform random points would over-represent empty rural space where nobody commutes and no cameras exist, understating exposure badly. Instead each origin–destination pair is drawn **with probability proportional to the number of workers actually making that trip**, so the distribution approximates what an average *commuter* experiences rather than what an average *square kilometre* contains.

Block-level LODES geocodes are aggregated to **tract** (first 11 digits), finer than the routing’s effective precision, and each tract is routed from its 2024 Census Gazetteer internal point [40] (INTPTLAT/INTPTLONG), which is guaranteed to fall inside the tract polygon. Same-tract commutes are dropped: origin and destination centroids coincide, so there is no route to measure. Georgia alone yields 1,199,250 distinct tract pairs covering 4,190,313 workers; across the 50 states and the District of Columbia the frame holds 132,822,679 commuters over 34,420,305 tract pairs.

Connecticut needed one repair. In 2022 it replaced its eight counties with nine planning regions, and every tract identifier changed with them. LODES still names tracts by the old counties while the 2024 Gazetteer and ACS 2022 use the new ones, so without a crosswalk no Connecticut tract matches and the state silently drops out of the sample. The tracts were reassigned, not redrawn: each six-digit tract code maps to exactly one new identifier, and all 879 in the flows do.

1,200 pairs are drawn per jurisdiction, 61,200 in all, with replacement from a single seeded stream (seed 20260725), giving **58,421 distinct commutes**. The fifteen states of the study’s earlier rounds are drawn first, so their 17,904 commutes are reproduced exactly and their routes carry over. The sampled pairs represent 2,842,747 workers; the median pair carries 16 and the largest 2,832.

A pair drawn more than once is routed once, since its route is the same, but it stands for every draw: **each commute carries its draw count as a weight**. In the fifteen large states this is immaterial—0.53% of draws are repeats—but in small ones it is not. Wyoming’s 1,200 draws hold 832 distinct pairs (30.7% repeats), Alaska’s repeat a quarter of the time and the Dakotas’ nearly a fifth, and one Vermont pair was drawn nine times. Collapsing repeats at equal weight, as this study’s earlier rounds did, under-represents precisely the heaviest flows; keeping the counts restores probability-proportional-to-workers sampling inside every state.

5 Method

5.1 Two routes per commute

Each sampled pair is routed twice against the same graph: a **baseline** with no custom model, and an **avoided** route applying `priority: [{if: "in_alpr", multiply_by: 0.01}]`. Both disable CH, which per-request custom models require.

Both use GraphHopper’s stock car model, which weighs distance as well as time (`distance_influence 90`: a route 1km shorter is preferred if it costs under 90s more). The baseline is therefore the router’s preferred route rather than strictly the fastest, and for 4.5 % of commuters the avoiding route—in every such case the longer one—is the quicker, by a median of 0.9 min. Extra time is reported as measured, negative values included; clipping them at zero leaves every median unchanged and raises the mean extra time from 5.62 min to 5.70 min.

0.01 is the application’s *strongest* avoidance setting, not a typical one: a watched road costs a hundred times its usual weight, close to but short of prohibition. Results therefore describe the **strong end of both the benefit and the cost** of avoidance as offered. This is deliberate: it establishes whether meaningful avoidance is possible at all, and what it costs near the limit.

5.2 Exposure is directional, not proximate

A camera counts as having seen the driver *only if the route line intersects its cone*. Passing 20 m from a camera aimed the other way is not exposure. Implementation is an STRtree over the 135,210 cone parts for candidate lookup, then an exact intersection test.

Two simplifications remain, and they pull in opposite directions: route geometry is a polyline while a vehicle has width, which understates exposure, and a cone occluded by intervening buildings is still counted as full sight-line, which overstates it.

5.3 Vendor attribution costs no additional routing

Routing does not change between vendor questions—the avoidance model targets the all-vendor union either way—so Section 6.4 is entirely a matter of re-scoring existing routes against different cone sets. This is possible only because the experiment persists route polylines to a sidecar; keeping only counts would have made both the vendor split and the radius sweep of Section 8.3 fresh multi-hour re-runs.

As a consistency check, the three vendor sets should sum to slightly *more* than the all-vendor union, because cones from different vendors that overlap merge into one part in the union and are counted once while the split sets count them separately. Measured excess is 1.0 %: right sign, plausible magnitude.

5.4 Earlier camera maps need no new baselines

The unavoided route does not depend on the cameras. Routed two months apart against graphs carrying different camera maps, all 17,580 commutes of the study’s earlier rounds took identical baseline routes, to the metre and the second. Exposure on an earlier date is therefore a re-scoring of each saved route against that date’s cones, with no routing.

Avoidance does need routing against that date’s cameras, and two properties keep it tractable. First, one graph serves every date: each date’s cone set is imported as its own custom area (`alpr_2025_07_01, ...`), and each request names the one it avoids—an import of about two

hours serves five or six dates rather than one, checked on a test graph to route each area independently. Second, only commutes that pass a camera on that date are routed. A route that crosses no cone is already the avoiding route, since the custom model only makes cone edges dearer and nothing can undercut it; this was checked rather than assumed, and all 3,741 such commutes of the fifteen-state run received an identical avoiding route. The same shortcut serves the main run.

Re-scoring uses route geometry stored at five decimal places, about a metre. Against live scoring on the current map it moves 1.7 % of commutes by one camera, in the two directions about equally, changes whether 5 of 17,580 (0.03 %) pass any camera at all, and leaves national totals within 0.1 %. On the full sample the re-scored national figures match the live ones to the precision reported (75.3 % passing a camera, a mean of 5.14). Within the trend every date, the current one included, is scored the same way. Exposure is re-scored on every monthly map; avoidance is routed on the first of every quarter, eleven dates from January 2024 to July 2026, and on the September 2026 snapshot itself. The 24 July 2026 snapshot enters re-scored only; the fifteen-state paired comparison covers it (Section 6.5).

5.5 Statistical reporting

- **Medians lead, means accompany.** Both distributions are heavily right-skewed; a minority of long commutes through dense corridors pull the mean well above the typical experience.
- **National figures are weighted by commuters.** Every jurisdiction was sampled at the same size, so an unweighted pool would count Wyoming’s 198,447 in-state commuters as heavily as California’s 16,821,903. Each routed commute carries its state’s commuter total from the LODES frame, shared among that state’s routed commutes in proportion to draw count, and quartile cut-points and medians are weighted quantiles. State figures are within-state estimates and need only the draw counts. The weights matter even on the fifteen states of the earlier rounds: weighting alone moves the share of commuters passing a reader there from 78.7 % to 81.1 %, because the populous states are the more surveilled ones.
- **Confidence intervals from a bootstrap stratified by state** [8] (percentile, 2,000 replicates, seed 20260725). Each replicate resamples every state at its own size and re-derives its weights, as the sample was drawn; a plain bootstrap over pooled rows would let one replicate hold 1,400 Texans and 900 Californians, variation the design does not have. Camera counts are discrete and zero-inflated, nowhere near normal, so no interval is assumed symmetric.
- **Commutes from one county are not independent.** They share corridors, and so cameras, and a bootstrap over commutes treats them as independent. Resampling whole origin counties within states instead [13] widens the intervals on the headline figures two- to threefold—75.3 % of commuters passing a reader becomes [74.0–76.4], the median extra time of 1.62 min [1.48–1.84]. On the sparse early maps, where a few counties hold most of the cameras, it matters more (Section 6.5); where the two bootstraps disagree, both are shown.

5.6 Execution

Twelve concurrent workers against the local routing server. Worker count was measured, not assumed: twelve completed 300 commutes in 324s, while thirty-two took longer than 576s on the identical set—past about twelve, contention outweighs added concurrency. Routing dominates cost overwhelmingly; the geometric exposure test is 0.2 % of per-commute work.

Failed routes are dropped rather than retried, and **56,131 of 58,421 commutes routed (3.9 % dropped)**. The failures are not spread evenly, and not mostly what they seem. Nine in ten are points the router cannot snap to any road: a tract’s internal point can lie deep inside a

Table 3: Exposure and the cost of avoidance for American commuters: 56,131 routed commutes in the 50 states and the District of Columbia, weighted to the 132.8 million in-state commuters they represent. Medians are weighted; intervals come from a bootstrap stratified by state. Median commute 23.6 km / 23.4 min.

	Median [95 % CI]	Mean
<i>Unavoided route</i>		
Cameras passed	3 [3–3]	5.14
Commuters passing ≥ 1	75.3 % [74.8–75.8]	
Commuters passing ≥ 5		36.5 %
Commuters passing ≥ 10		16.3 %
90 th / 99 th percentile	13 / 34 cameras	
Exposure rate	0.096 cameras/km	
<i>ALPR-avoiding route</i>		
Cameras passed	0 [0–0]	0.26
Reduced to zero	84.2 % [83.7–84.7]	
Extra time (min)	1.62 [1.57–1.68]	5.62
Extra distance (km)	0.57 [0.55–0.60]	2.53
Avoidance overhead	6.71 % [6.50–6.93]	13.27 %
Minutes per camera evaded	0.76 [0.75–0.78]	1.30

national park, a reservation or a wilderness tract, beyond the router’s road search. The rest are genuinely unconnected—commutes between Hawaiian islands, or to Alaskan towns off the road network. Five states lose more than a tenth of their sample (Alaska 33 %, Hawaii 24 %, Montana and Wyoming 21 %, New Mexico 14 %); the median state loses 2.0 %. Those five hold 1.6 % of the nation’s commuters, so the national figures barely register the loss, but their own estimates describe the routable part of each state—probably its less remote part, and so probably overstate its exposure. A wider road search would recover most of them, but it also moved 5 of 100 already-routed commutes to a different road (the default search sometimes stops at a road that is not the nearest), and the trend of Section 6.5 pairs every saved route with new ones. Consistency was kept over coverage.

Commutes are processed in seeded random order. The sample is grouped by state and the executor preserves input order, so in file order results arrive one state at a time: an interruption would leave some states complete and others absent, and no partial output could support a national estimate. Shuffling makes any prefix a near-uniform sample of the whole. This proved material rather than hypothetical—in the study’s first round an interim read of a single state gave a median avoidance overhead of 22.7 % against that round’s national 5.96 %.

6 Results

6.1 Exposure on the unavoided commute

75.3 % of American commuters pass at least one ALPR on the way to work. The median commuter passes three, the 90th percentile thirteen and the 99th thirty-four. Aggregate exposure is 0.096 cameras per kilometre driven. Across the fifteen states of the study’s earlier rounds the same weighted share is 81.1 %: the thirty-five states and the District added here are, on the map, far less surveilled—a difference Section 6.3 will not let us attribute to the roads rather than to the map.

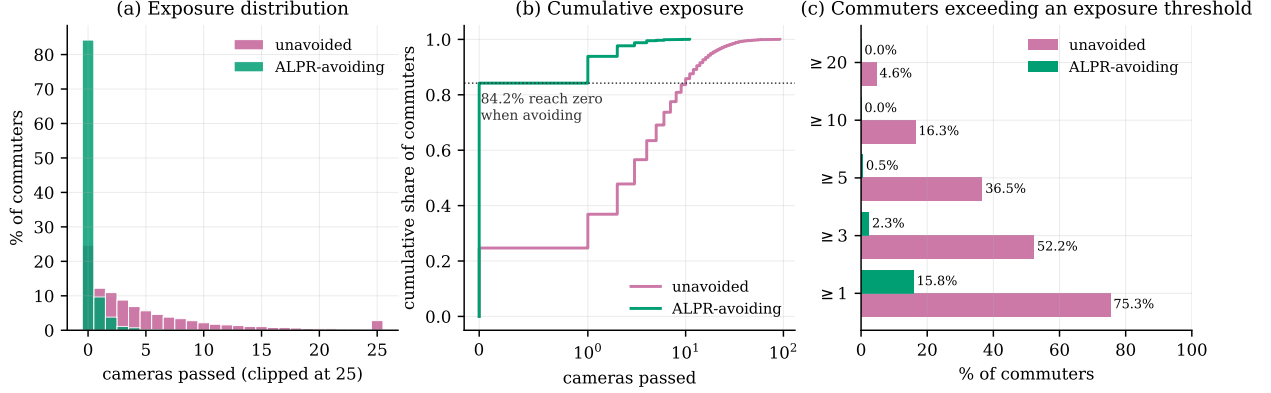


Figure 2: Exposure before and after ALPR-avoiding routing, as shares of commuters. (a) Distributions; (b) cumulative, log-scaled abscissa; (c) share of commuters exceeding a threshold. The zero-exposure mass moves from 24.7% to 84.2%.

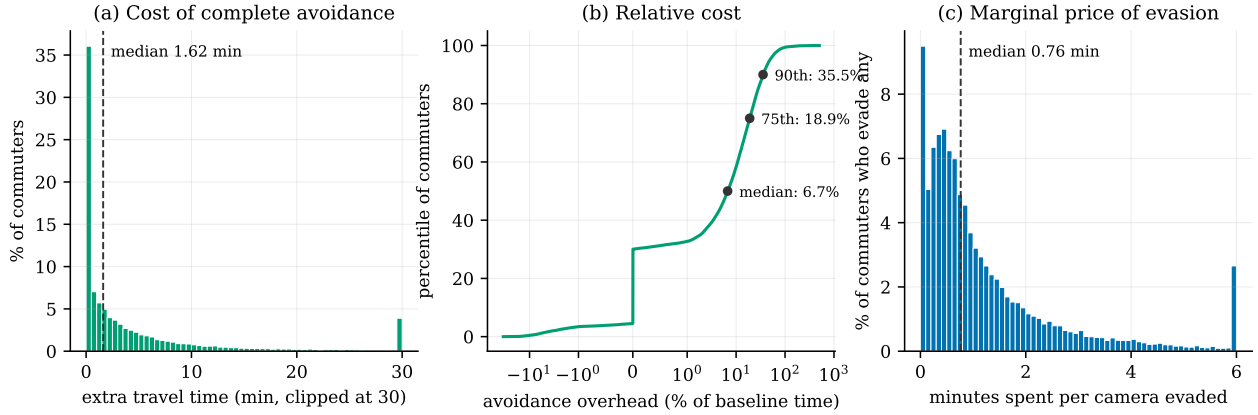


Figure 3: The price of complete avoidance, as shares of commuters. (a) Absolute extra time; (b) overhead as a percentage of baseline, by percentile; (c) marginal minutes per camera evaded. All three are heavily right-skewed—a small tail of commutes pays a great deal, while the typical commute pays almost nothing.

6.2 The cost of refusal

84.2% of American commuters could be routed to zero ALPR exposure: the quarter whose usual route passes no mapped reader, and four in five of the rest (79.0% [78.4–79.7], Table 5). Over all commuters the avoiding route adds a median of 1.62 min and 0.57 km, an overhead of 6.71%; for commuters whose usual route passes a camera it adds a median of 3.14 min [3.05–3.22]. The median marginal price is 0.76 min per camera evaded.

For most commuters, then, refusing licence-plate surveillance on the way to work costs a few minutes at most. This is conditional on the assumed camera field of view: it is stable if the true radius is smaller than the 60 m assumed here and degrades if it is substantially larger, which Section 8.3 quantifies. It is the paper’s central quantitative claim.

The distribution matters as much as the median. The mean overhead is 13.27% against a median of 6.71%, and the interquartile range runs from 0% to 18.9%: a quarter of commuters need no detour whatsoever, while a tail pays substantially. Figure 4 shows one such case.

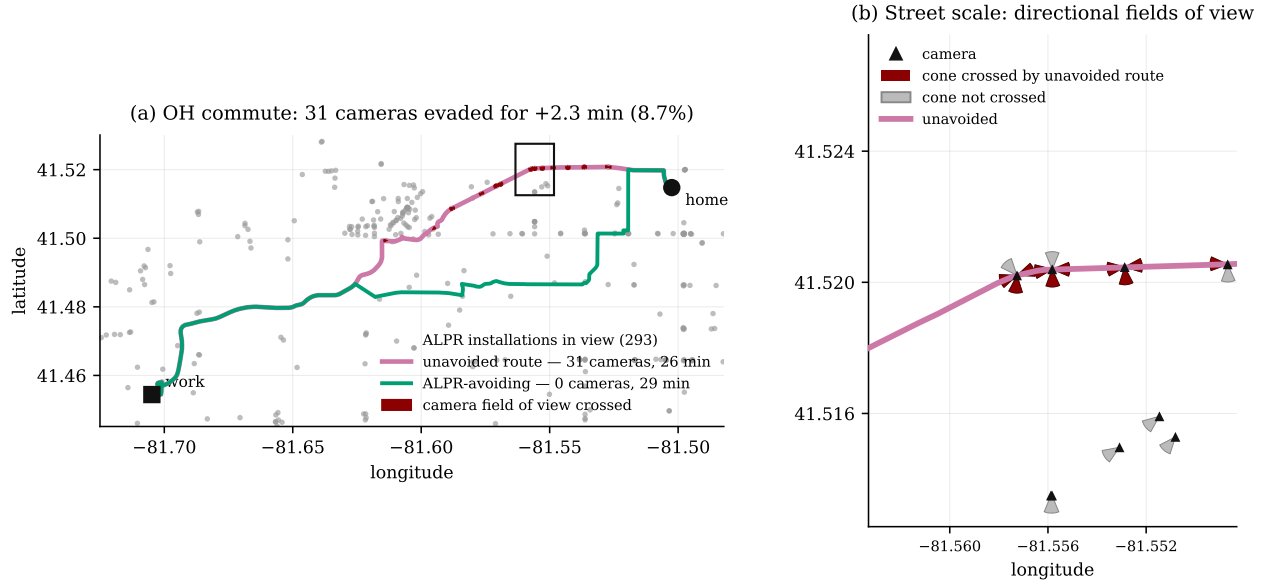


Figure 4: An illustrative high-exposure commute in the Cleveland area—chosen for legibility, not typicality. (a) The unavaoided route crosses 31 camera fields of view; the avoiding route crosses none, for 2.3 min (8.7%). (b) At street scale the directional model is visible: cones the unavaoided route crosses are dark, cones aimed elsewhere are grey and cost nothing. Map data © OpenStreetMap contributors.

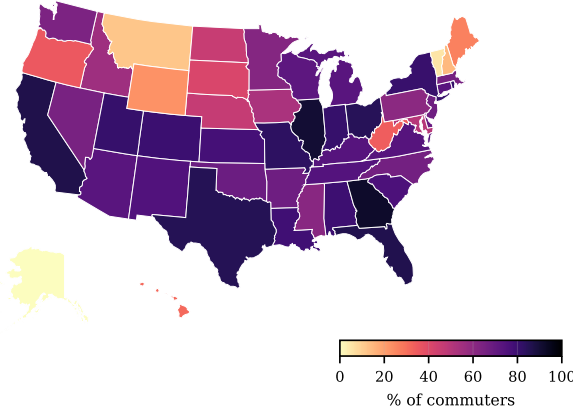
6.3 Fifty states and the District of Columbia

State-level variation is extreme (Figure 5; every state is tabulated in Table 7). In Georgia 91.9% of commuters pass a mapped reader, a mean of 11.8 on the way to work, and avoiding them costs a median 25.7% overhead; in Illinois 90.6% do, in California 86.7%. At the other end 14.8% of New Hampshire’s commuters pass one, 11.8% of Montana’s, 5.4% of Vermont’s, and none of Alaska’s sampled commuters. In fifteen jurisdictions the median extra time is zero: most of their commuters have nothing on the map to avoid.

These are differences in the map at least as much as on the roads. Mapped density runs from 0.7 cameras per 100,000 residents in Alaska, and under five in Vermont, Maine and New Hampshire, to 92.8 in Georgia—a 7.1-fold spread between the 10th and 90th percentile states—and state exposure tracks it at $r = 0.82$. The departures from the line are what a statewide rate would produce: New York’s commuters pass more cameras than its statewide density predicts, because commuting concentrates where cameras do while the density is averaged over the whole state; Kansas’s pass fewer.

Per camera, evasion tends to be cheaper where cameras are denser. Across the 48 states with at least 100 evading commuters, the median price per camera falls as cameras per kilometre rise ($r = -0.29$ against the logarithm of cameras per kilometre, 95% CI -0.53 to -0.01), from 2.49 min in New Hampshire to 0.17 min in the District of Columbia. Dense grids offer a parallel street one block over; rural and intercity routes often have one road. Across states the relationship is weak—Georgia, the most densely watched large state, still pays 1.11 min a camera—so we state it as a tendency, not a law: the burden of refusal falls, if anything, hardest on drivers with the fewest cameras to refuse.

(a) Share of commuters passing at least one ALPR



(b) Measured exposure tracks mapping intensity

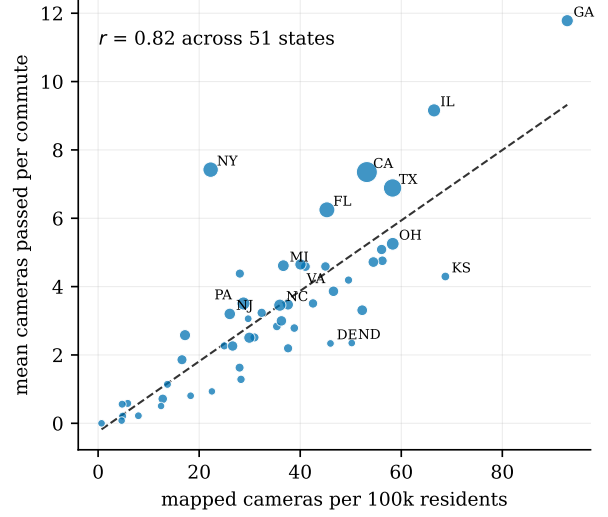


Figure 5: (a) Share of each state’s commuters whose unavoids route passes at least one mapped ALPR. (b) Mean cameras passed against mapped cameras per 100,000 residents, points sized by commuters. Mapping intensity alone tracks most of the variation between states ($r = 0.82$): from a single map, a state with few mapped cameras cannot be told from a state with few mappers (Section 8.2).

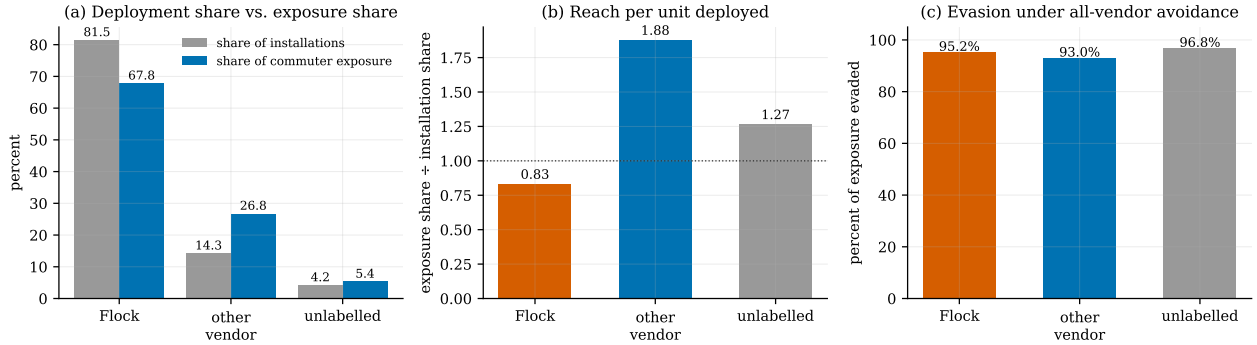


Figure 6: Vendor-resolved exposure, commuter-weighted. (a) Flock’s share of exposure falls well below its share of installations. (b) Reach per unit deployed: share of exposure divided by share of installations. (c) Both are avoided at comparable rates under the all-vendor model. Installations are the cameras that contribute cones.

6.4 By camera vendor

Flock Safety holds 81.5 % of the mapped installations that contribute cones in the fifty states and the District, but supplies only 67.8 % of commuter exposure. Per unit deployed, non-Flock cameras generate **2.26** $\times$ the exposure (Table 4). Two differences in how and where they stand go with it. Non-Flock units are multi-head more than six times as often (Table 2), and they stand on busier roads: 42 % of them within 30 m of an Interstate, freeway or other principal arterial in the federal traffic inventory [11], against 26 % of Flock’s, and 30 % on roads OpenStreetMap classes as primary, against 22 %. A Flock unit more often watches a single approach on a road fewer commutes cross.

The difference is not one of customer. Where the operator is tagged—for a quarter of the other

Table 4: Vendor-resolved exposure, commuter-weighted over the 51-state sample. Installation shares are of the 139,391 cameras in the 50 states and the District of Columbia that contribute cones. Reach is share of exposure divided by share of installations: Flock holds four-fifths of installations and supplies two-thirds of exposure, and per unit deployed a non-Flock camera generates $2.26\times$ as many camera encounters.

Set	Installations	Share of exposure	Reach	Commuters ≥ 1	Evaded
Flock Safety	81.5 %	67.8 %	0.83	65.5 %	95.2 %
Other vendor	14.3 %	26.8 %	1.88	32.0 %	93.0 %
No vendor tag	4.2 %	5.4 %	1.27	11.6 %	96.8 %

Table 5: The same 56,131 commutes on the same roads against the camera map of each routed date, commuter-weighted. Exposure from re-scoring each saved route; avoidance from routing every commute exposed on that date, with the share reaching zero and the median extra time given for those commuters. The monthly maps between these dates are in Figure 7.

Map date	Mapped ALPR	Passing ≥ 1 (%)		Mean passed	Exposed who reach zero (%)	Extra min, exposed	Min per camera
			[95% CI]				
2024-01-01	1,112	2.2	[2.0–2.4]	0.05	95.0	0.77	0.42
2024-04-01	1,267	2.9	[2.7–3.1]	0.08	95.7	1.04	0.51
2024-07-01	1,420	3.3	[3.0–3.5]	0.08	96.0	1.00	0.49
2024-10-01	1,581	3.5	[3.2–3.7]	0.09	96.0	1.02	0.52
2025-01-01	7,583	12.3	[11.9–12.7]	0.35	92.5	1.20	0.61
2025-04-01	14,238	20.4	[19.9–20.9]	0.59	92.7	1.24	0.63
2025-07-01	22,690	29.4	[28.8–30.0]	0.94	92.4	1.36	0.65
2025-10-01	34,383	38.4	[37.8–39.0]	1.33	91.6	1.51	0.66
2026-01-01	63,522	55.9	[55.3–56.4]	2.44	86.9	1.98	0.70
2026-04-01	85,845	63.4	[62.8–63.9]	3.14	84.7	2.32	0.73
2026-07-01	109,853	69.6	[69.1–70.1]	4.08	82.6	2.67	0.75
2026-09-22	142,257	75.3	[74.8–75.7]	5.14	79.0	3.14	0.76

vendors’ cameras and an eighth of Flock’s—police departments operate about two-thirds of each (64 % and 67 %); transport agencies operate 15 % of the other vendors’ tagged cameras and almost none of Flock’s, whose second customer is retail business (21 %). The other vendors are chiefly Motorola Solutions, Genetec, Axis and Leonardo. Flock wins on count and loses on reach because of where its cameras stand—not, as far as operator tags show, because of who buys them.

6.5 Thirty-three months of mapping

Everything above describes the map of 22 September 2026. We re-asked the same questions of the map as it stood on the first of every month from January 2024, reconstructed exactly from OpenStreetMap’s history (Section 4.3), and of the July 2026 snapshot: the same 56,131 commutes on the same roads, each saved route re-scored against each month’s cones, and, on the first of every quarter, every commute that passed a camera on that date routed again to avoid that date’s cameras (Section 5.4). Table 5 gives the routed dates, Figure 7 every month, and Figure 8 every state. Each date should be read as what had been *mapped* by then, not what had been installed.

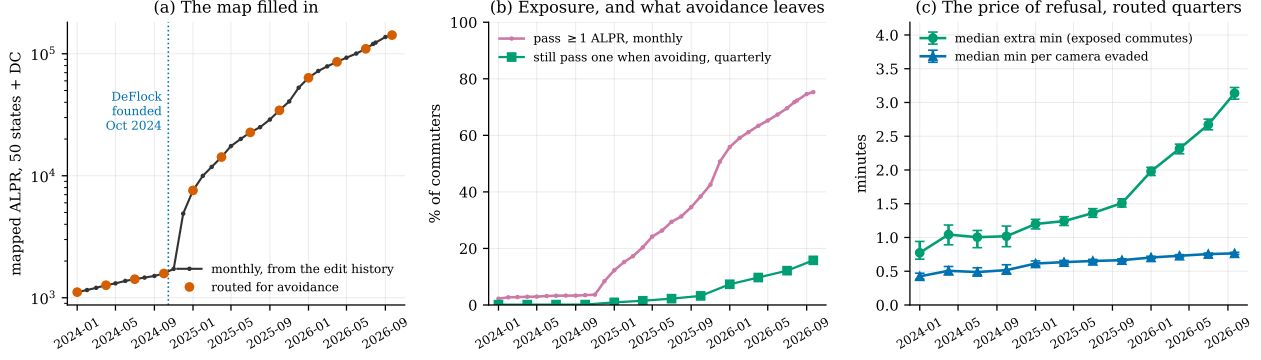


Figure 7: (a) Mapped ALPR in the fifty states and DC on the first of every month, from the edit history, with the routed dates marked. (b) Share of commuters passing at least one mapped reader on each monthly map (band: 95 % CI), and the share still passing one on the avoiding route, on the routed dates. (c) For commuters exposed on each routed date, the median extra minutes to avoid every camera, and the median minutes per camera evaded.

The map filled in. On 1 January 2024 OpenStreetMap held 1,112 ALPR installations in the fifty states and the District of Columbia, four-fifths of them with no vendor tag, and twenty-six jurisdictions had none at all. It grew by about fifty a month until DeFlock was founded in October 2024, and then abruptly: 1,724 on 1 November, 4,894 on 1 December, 7,583 on 1 January 2025. From there the map grew by a median 10.7 % a month—22,690 by July 2025, 63,522 by January 2026, 109,853 by July, and 142,257 on 22 September 2026, a 128-fold increase in thirty-three months (Figure 7a)—and the pace has not slackened: the months of 2026 added a median 9.4 % each. By July 2025 only Hawaii, Vermont and Alaska had no mapped reader; by January 2026 only Alaska; by July 2026 none. Flock’s share of the mapped network went from 17 % to 72 % during 2024 and has stayed near four-fifths since, while the untagged share fell from 81 % to 5 % and the named non-Flock vendors rose from 1 % to 15 % as contributors tagged them.

Exposure followed the map, almost in proportion. The share of commuters passing at least one mapped reader crossed a tenth in January 2025, a quarter in June, a half in December, and stands at three-quarters (75.3 %) today; the mean went from 0.05 cameras to 5.14. Against the number of mapped cameras the elasticity is 0.89 across the thirty-five maps. Per camera mapped, exposure has fallen by about a quarter since January 2024, from 0.049 to 0.036 encounters per commute per thousand cameras: the cameras mapped later lie on commute routes slightly less often than those mapped first, but not much less. Nor is there a sign of saturation. The two months between the July and September 2026 snapshots alone brought 18.3 % more mapped cameras and 17.0 % more encounters, and the monthly curve of Figure 7b is still rising by about two points a month.

The states filled in at different times. In January 2024 Illinois was the only state where more than a tenth of commuters passed a mapped reader (23.5 %); by January 2025 fifteen were, by July 2025 thirty-one, by January 2026 forty-seven, and today forty-nine (Figure 8). In July 2025 Michigan’s commuters were the most exposed of any state’s (62.8 %), while Pennsylvania (2.4 %), Nevada (3.8 %), New Jersey (4.4 %) and North Carolina (8.2 %) would have looked nearly camera-free; all four are between 60 % and 68 % today. A state’s rank on the January 2025 map predicts its rank today only moderately (Spearman $\rho = 0.79$ on mean exposure; from January 2024, 0.43;

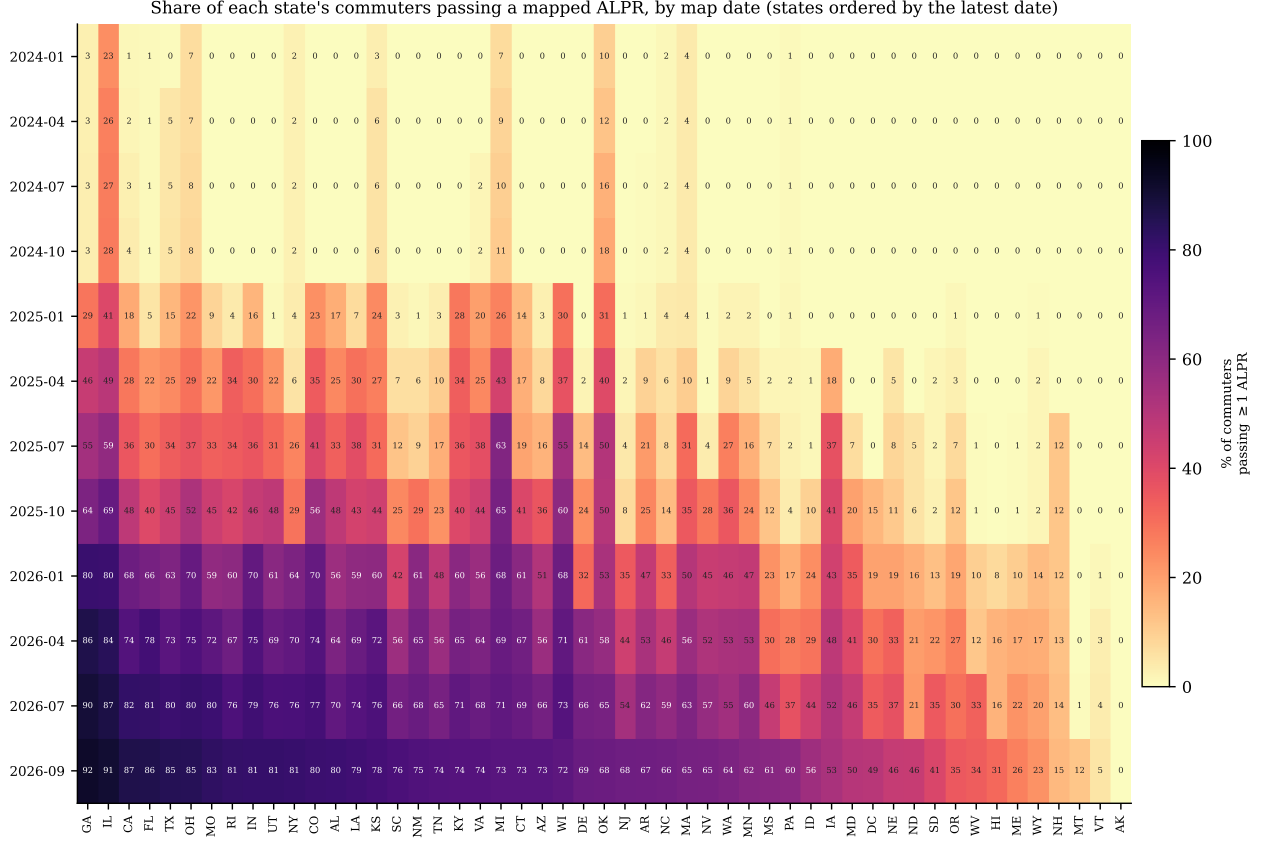


Figure 8: Share of each state's commuters passing at least one mapped ALPR on each routed map, states ordered by the latest.

from January 2026, 0.90). Nor does the map only grow. Wisconsin and Iowa each gained cameras on net between July and September 2026 yet lost about three points of exposure: 190 and 123 of their July cameras left the map, deleted or retagged, and the cameras newly mapped added less exposure than was lost.

Refusal got dearer, and not only because there was more to refuse. On the sparse maps of 2024 a commuter who passed a mapped camera could almost always get around it: 95.0% [92.7–96.9] of exposed commuters could reach zero, for a median of 0.77 min. On the September 2026 map 79.0% [78.4–79.7] can, for a median of 3.14 min (Table 5; Figure 7c). The quarterly maps show the shape of the change. The share reaching zero held at 95% to 96% through 2024, stepped down to about 92% with the surge of mapping that followed DeFlock's founding and held there through 2025, and has fallen on every routed map since October 2025: 91.6%, 86.9%, 84.7%, 82.6% and 79.0%, each interval clear of the last. The cost did not step; it climbed. From July 2024 on, the median extra time for an exposed commuter rose on every routed map, from 1.00 min to 3.14 min. Most of that rise is the number of cameras on each trip, but the price of evading each one rose too, from 0.42 min [0.38–0.47] to 0.76 min [0.75–0.78], and it too rose on every routed map from July 2024. Two things moved it, and a fixed sample can separate them. One is who was exposed: the early maps covered dense metropolitan grids, where Section 6.3 finds evasion cheapest, and the later ones reached rural and intercity roads. The other happened on the same trips. Followed from the July 2025 map to September 2026's, the 12,647 commutes exposed in July came to pass 9.9

cameras on average instead of 3.2; the share of them that can reach zero fell from 92.4 % to 72.6 %, and the price per camera rose from 0.65 min [0.63–0.68] to 0.81 min [0.79–0.84]. Every cohort shows it, and for about a year the rise grows with the time a cohort is followed: the commutes exposed on the July 2026 map pay 2 % more per camera twelve weeks later, too little to distinguish; April 2026’s pay 7 % more, January’s 11 % and October 2025’s 20 %, each interval clear of its starting one. The seven cohorts followed for fourteen months or more pay 21 % to 35 % more, six of them clear of their starting intervals and the seventh, April 2024’s, overlapping it slightly. As more of a trip’s cameras, and of its alternatives’, are mapped, each costs more to avoid. Over the two months between the 2026 snapshots the change is too small to see, which is why the fifteen-state comparison below finds the price per camera unchanged.

Fifteen states, fully re-routed. The trend’s last step can be checked against a comparison that re-routed everything. For the fifteen states of the study’s earlier rounds, the same 17,580 commutes were routed in full against both 2026 snapshots on the same road extract, and every baseline route was identical to the metre and the second. Camera encounters grew by 18.3 % against 18.7 % more mapped cameras in those states, the median price of evading one stayed at 0.78 min to the second decimal, and Pennsylvania’s mean exposure rose 88 % in two months.

What follows. Two things. The absolute exposure figures of this paper are a lower bound that has risen with every map and will keep rising; any study built on crowdsourced surveillance data dates, and should say when its map was drawn. And the price of refusal dates as well: it has risen with every map, per camera as well as in total, and today’s figures describe a map that is still filling in.

7 Discussion

7.1 Refusal is cheap, and that is the policy-relevant fact

The strongest result here is also the simplest: 84.2 % of American commuters can reach work past no mapped licence-plate reader, and the avoiding route is a median of 1.62 min longer—3.14 min for commuters with a camera to avoid. A common intuition about these networks is that they are, or soon will be, comprehensive enough that avoidance is futile, so that the privacy loss is unavoidable rather than chosen. At current mapped density that is false for five commuters in six. It has been false on every map since 2024, but it is becoming less so: as the map filled in, the share of exposed commuters who can reach zero fell from 95 % to 79 %, and the median cost for those with a camera to avoid quadrupled (Section 6.5).

The corollary cuts the other way. If avoidance is this cheap, it is also cheap for people with reason to evade legitimate investigation, which bounds how much investigative value a fixed-camera network can deliver against a motivated subject while it continues to read every unmotivated one.

7.2 Attention and exposure are aimed at different infrastructure

Flock Safety is the name in municipal council debates, in local journalism, and in the activist mapping effort that produced the data underlying this paper. It holds four-fifths of mapped installations. Yet per unit deployed its cameras generate fewer than half as many camera encounters as other vendors’.

We do not read this as exonerating, for a reason the measurement itself makes clear. A camera at the entrance of a residential subdivision does something qualitatively different from a multi-head

unit over a six-lane arterial: it observes a small population repeatedly rather than a large population once, and repeated observation of a known small set is precisely what makes pattern-of-life inference possible. Commute exposure measures breadth, not depth, and Flock’s model appears optimised for depth. A study designed around repeat observation of the same vehicle would likely reverse the ordering reported here, and we would expect it to.

What the result does establish is that public attention and measured exposure are attached to different infrastructure: per unit, the cameras commuters pass most are other vendors’, bought, where the operator is known, mostly by police departments too, but standing beside the arterials and freeways that commuters share. Regulation aimed only at the vendor in the news would leave most of what a commute passes per unit untouched.

7.3 Where the burden actually concentrates

The distributional finding that rests least on the map is geographic, and it is counter-intuitive: per camera, avoidance tends to be cheapest in dense metropolitan grids and dearest in rural and intercity travel. Network topology, not camera count alone, sets the price of refusal. The tendency is modest across whole states (Section 6.3), but it rests on road-network structure—which OSM maps close to completely—rather than on camera counts, so it is robust to the coverage critique in a way no result built on counts can fully be, however well checked.

8 Limitations

8.1 Tract centroids are not addresses

Every route runs centroid to centroid; real commutes begin in a driveway. In large or irregular tracts the centroid can sit far from any residence. This adds noise in both directions and is the main reason we report distributions rather than point estimates. It also costs coverage: where a tract’s internal point lies too far from any road to snap to, the commute cannot be routed at all, which removes a fifth or more of the sample in Alaska, Hawaii, Montana and Wyoming (Section 5.6).

8.2 Crowdsourced camera data is non-randomly missing

Absolute exposure figures are a **lower bound**: like volunteered geographic data generally [21, 22], the map is least complete where contributors are absent, and there cameras are missing from the map but present on the pole. Two checks bound it. Against Flock Safety’s own transparency portals the map holds, at the median, 0.98 of the cameras that 1,050 municipal police departments report operating (28,349 mapped inside their limits against 31,210 reported), small towns being the least completely mapped [10]. Against national totals it is less complete: Flock has said it operates more than 120,000 cameras nationwide, and a researcher’s map built from its records in December 2025 shows more than 170,000 devices of all kinds [29], against the 114,352 Flock cameras on our map in the fifty states and DC—so the map holds at most 95 %, and perhaps two-thirds, of Flock’s cameras. Coverage outside the portal cities, and of cameras no department reports, is the larger unknown, and the trend of Section 6.5 is the best evidence of how far the map still has to go.

8.3 Cone geometry: swept, and one result depends on it

The 60 m radius was chosen by reasoning—a pole set back from the kerb needs a cone long enough to reach the carriageway it watches—and the 45° default span from the 82 % of explicit arcs that

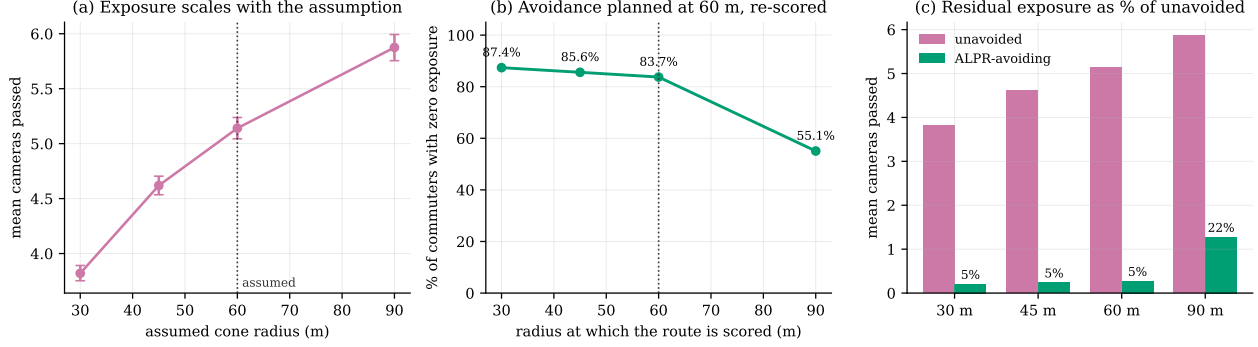


Figure 9: Sensitivity to the assumed cone radius. (a) Measured exposure scales gently— halving the radius removes only a quarter of it. (b) The share of commuters reaching zero exposure is flat below the assumed value and falls sharply above it. (c) Residual exposure as a fraction of unavoided, by radius.

span exactly that. Neither was measured. We therefore swept the radius over $\{30, 45, 60, 90\}$ m, rebuilding the cone geometry at each and re-scoring all 56,131 routes (Figure 9).

Table 6: Cone-radius sensitivity, commuter-weighted. Exposure is mildly sensitive to the assumption; the avoidance result is not, until the radius grows by half. At the assumed 60 m the clean share is 83.7 % rather than the 84.2 % of Table 3 because the sweep re-scores stored route geometry (Section 5.4).

Radius	Mean cameras	vs 60 m	Avoided routes still clean	Residual
30 m	3.82	0.74×	87.4 %	5.4 %
45 m	4.62	0.90×	85.6 %	5.2 %
60 m	5.14	assumed	83.7 %	5.3 %
90 m	5.88	1.14×	55.1 %	21.7 %

Two things follow, and they point in opposite directions.

Exposure is robust. Halving the radius removes only a quarter of the measured exposure, and 90 m adds 14 %. Cones are small relative to the road network at any of these values, so Section 6.1 does not rest on the choice.

The avoidance headline does depend on it. Routes planned against 60 m cones remain clean when re-scored at 45 m and 30 m—avoidance is not finely tuned to the assumption from below. But at 90 m the share reaching zero falls from 83.7 % to 55.1 %, a drop of 28.7 points (95 % CI $[-29.3, -28.1]$), and residual exposure quadruples from 5.3 % to 21.7 % of baseline. If real fields of view are half again as long as we assume, more than a third of the routes this paper counts as clean are not.

That figure is nonetheless a *lower* bound rather than an estimate, for a reason worth being precise about. These routes were *planned* against 60 m cones and only *re-scored* at 90 m; a router given 90 m cones would find different clean routes, and the denser the network the more of them there are. Measuring the true value requires a complete re-route at each radius—most of a day apiece at this sample’s size—which we have not done. What the sweep establishes is that the avoidance claim is conditional on the radius in a way the exposure claim is not, and that the conditionality bites only above the assumed value.

8.4 A single avoidance strength

All results use the strongest setting. Intermediate settings are the practically interesting ones for a shipped application, and in informal tests they behaved badly: the useful range was roughly $0.3 \rightarrow 0.01$, and whether a setting took effect depended on local camera density. On two test routes a setting of 0.3 avoided none of six cameras between Wilmington and Dover, Delaware, and five of eight across metropolitan Atlanta. A single fixed slider-to-multiplier mapping is therefore unlikely to serve; the mapping may need to adapt to local density. This is an open problem for the application and an unmeasured dimension of this paper.

8.5 Sample scope

All fifty states and the District of Columbia, but within-state commutes only (`od_main`), so cross-border metropolitan areas—New York/New Jersey, Philadelphia, Washington, Kansas City, St Louis, Portland—are systematically truncated at the state line. Several are exactly the dense multi-jurisdiction areas where the question is most pointed. The District is the extreme case: most people who work there live in Maryland or Virginia, so its sample describes only commutes with both ends inside the District. Alaska’s flows are from 2016 and Michigan’s from 2021, the latest LODES publishes; the other 49 jurisdictions’ are from 2022.

8.6 Past maps record mapping, not deployment

The trend of Section 6.5 is a trend in what had been mapped. The edit history gives the day each camera entered the map to the second; it does not give the day the camera went up, which OpenStreetMap records for 47 of 142,991 nodes. A camera first mapped in 2026 may have stood since 2021, and the 128-fold growth of the map cannot be read as a rate of deployment. Absolute figures date accordingly. The relative claims are the ones to carry forward, and they held on every map: the cheapness of refusal—under a minute per camera evaded on all twelve routed maps, though rising on each—and the near-proportionality of measured exposure to the map across all thirty-five. What the maps cannot establish is that the current one is complete.

8.7 Free-flow travel times

Routing uses free-flow speeds with no traffic, time-of-day, or turn-restriction penalties, so both travel times are optimistic. The *difference* between them—the quantity actually reported—is more robust than either absolute, since both share the bias.

8.8 Avoidance is a preference, not a guarantee

`multiply_by`: 0.01 makes observed roads expensive, not impassable. Where no alternative exists the router correctly uses the watched road. “Zero cameras” means the optimiser found a clean path, not that one was guaranteed to exist.

9 Ethics and dual use

The study uses only public data. The camera locations are volunteers’ observations from public places, already published in OpenStreetMap under an open licence; the commute flows and neighbourhood attributes are Census Bureau aggregates, published with the Bureau’s confidentiality protections and attached to census tracts rather than to people; the portal counts are published by

the police departments themselves. No individual is identified, no one’s travel was observed, and the paper adds no camera location to the public record.

The routing engine is dual-use, and we weighed that before publishing it. It lets anyone decline to be photographed on the way to work, including someone with reason to evade a lawful investigation. Three considerations decided the matter. The capability is not new: the camera map is public, and several services already route drivers around it [1, 5, 15]. The result it rests on bounds the investigative value of fixed cameras against a motivated subject whether or not the tool exists (Section 7); what such a network reliably records is the unmotivated majority. And the measurement itself—how much surveillance an ordinary trip passes, what declining it costs, and how fast both are growing—is information a public debate about the technology needs and has not had. The software is released under GPLv3, and its routing server is not exposed to the internet by default (Section 3).

10 Reproducibility

Code, configuration, and the derived outputs behind every table and figure—per-commute results, per-state tables, the siting tables and every analysis report quoted in the text—are in the project repository, <https://github.com/KaraZajac/TRAMES>, under `research/`. Fixed seed 20260725 throughout, for both sampling and bootstrap. All external data sources are public and require no API key. Beyond the routing server, the analysis needs Python with NumPy, Shapely and Matplotlib; pyosmium for the road sample and the history dump; and pyogrio, which bundles GDAL, only to read the HPMS geodatabase.

The 18 GB OSM extract, the 14 GB graph caches, the 163 GB OSM full-history dump, the 1.2 GB of TIGER/Line boundaries and the 5.3 GB HPMS geodatabase are not redistributed; the acquisition and import scripts rebuild all of them. The transparency-portal counts behind the coverage check come from a live service and are kept as retrieved, in `research/data/validation/`. Commuter weights come from `research/out/sampling_frame.json` and `sample_draws.csv`, written by the sampler.

References

- [1] BanditKin. Flock camera map: Find & avoid ALPR cameras near you, 2026. URL <https://banditkin.com/flock-camera-map>. Accessed 2026-09-24.
- [2] Sarah Brayne. Big data surveillance: The case of policing. *American Sociological Review*, 82 (5):977–1008, 2017. doi: 10.1177/0003122417725865.
- [3] Carpenter. *Carpenter v. United States*, 585 U.S. 296. Supreme Court of the United States, 2018. Decided 22 June 2018.
- [4] DeFlock. DeFlock: Mapping automated license plate readers, 2024. URL <https://deflockk.org>. Founded October 2024 (<https://deflockilm.org/what-is-deflock/>); formerly deflock.me; accessed 2026-09-25.
- [5] DeFlock. DeFlock maps: ALPR camera map and privacy routes, 2026. URL <https://maps.deflock.org>. Accessed 2026-09-25.
- [6] Drivers Against Flock. Private routes around ALPR cameras, 2026. URL <https://driversagainstflock.org>. Accessed 2026-09-24.

Component	Path
Data acquisition	research/data/fetch.sh
Camera fetch (continental)	server/alpr/fetch-na-cones.sh
Cone construction	server/alpr/build_cones.py
Past camera maps (Overpass)	server/alpr/fetch-history.sh
Past camera maps (history dump)	server/alpr/history_from_planet.py
Every version of every camera node	server/alpr/node_dates.py
Camera map at any instant	server/alpr/snapshot_from_versions.py
Merged device map, all sources	research/scripts/build_supermap.py
Roads each device watches	research/scripts/roads_watched.py
Cone-set variants	server/alpr/build-variants.sh
Snapshot statistics	research/scripts/snapshot_stats.py
Routing configuration	server/graphhopper/config/trames.yml
Graph build	server/graphhopper/rebuild-with-cones.sh
Commute sampling	research/scripts/build_sample.py
Experiment	research/scripts/run_experiment.py
Routing on past maps	research/scripts/run-history.sh
Commuter weights, bootstrap	research/scripts/wstats.py
County-cluster bootstrap	research/scripts/cluster_check.py
Trend	research/scripts/trend.py
Road-point sample	research/scripts/road_sample.py
Boundaries, projection	research/scripts/geo.py
Traffic sample (HPMS)	research/scripts/hpms_sample.py
Vendor road classes, operators	research/scripts/siting_checks.py
Coverage against portals	research/scripts/validate_coverage.py
Commute-mode reweighting	research/scripts/commute_mode.py
Geometry backfill	research/scripts/backfill_geometry.py
Vendor rescoring	research/scripts/vendor_exposure.py
Analysis	research/scripts/analyze.py
Post-processing, one command	research/scripts/run-analyses.sh
Snapshot comparison	research/scripts/compare_snapshots.py
Figures	research/paper/make_figures.py
Tables	research/paper/make_tables.py

- [7] Anne Edmundson, Roya Ensafi, Nick Feamster, and Jennifer Rexford. Characterizing and avoiding routing detours through surveillance states. arXiv:1605.07685, 2016. URL <https://arxiv.org/abs/1605.07685>.
- [8] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Number 57 in Monographs on Statistics and Applied Probability. Chapman & Hall, New York, 1993.
- [9] Electronic Frontier Foundation and Reynolds School of Journalism, University of Nevada, Reno. Atlas of Surveillance, 2026. URL <https://atlasofsurveillance.org>. Accessed 2026-09-24.
- [10] Eyes On Flock. Eyes on flock: Aggregating Flock Safety transparency portal data, 2026. URL <https://eyesonflock.com>. From <https://transparency.flocksafety.com>; retrieved 2026-09-24, snapshot in research/data/validation/.
- [11] Federal Highway Administration. Highway performance monitoring system (HPMS) 2024. U.S. Department of Transportation, Bureau of Transportation Statistics, National Transportation Atlas Database, 2025. URL <https://geodata.bts.gov/datasets/5e6a977c2d7c4ec1bdc82e684d3384f2>. Compiled 23 December 2025; file geodatabase retrieved 2026-09-24.

- [12] Andrew Guthrie Ferguson. *The Rise of Big Data Policing: Surveillance, Race, and the Future of Law Enforcement*. New York University Press, New York, 2017.
- [13] Christopher A. Field and Alan H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.
- [14] Kristin Finklea. Law enforcement and technology: Use of automated license plate readers. Technical Report R48160, Congressional Research Service, August 2024. URL <https://www.congress.gov/crs-product/R48160>.
- [15] FlockHopper. FlockHopper: Flock camera map & routes that avoid them, 2026. URL <https://dontgetflocked.com>. Accessed 2026-09-24.
- [16] Robert Geisberger, Peter Sanders, Dominik Schultes, and Daniel Delling. Contraction hierarchies: Faster and simpler hierarchical routing in road networks. In Catherine C. McGeoch, editor, *Experimental Algorithms: 7th International Workshop, WEA 2008*, volume 5038 of *Lecture Notes in Computer Science*, pages 319–333, Berlin, Heidelberg, 2008. Springer. doi: 10.1007/978-3-540-68552-4_24.
- [17] Geofabrik GmbH. OpenStreetMap data extracts: North america, 2026. URL <https://download.geofabrik.de/north-america.html>. Retrieved 2026-07-24; data © OpenStreetMap contributors, Open Database License 1.0.
- [18] Andrew V. Goldberg and Chris Harrelson. Computing the shortest path: A* search meets graph theory. In *Proceedings of the Sixteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA '05)*, pages 156–165, Philadelphia, 2005. SIAM.
- [19] Michael F. Goodchild. Citizens as sensors: The world of volunteered geography. *GeoJournal*, 69(4):211–221, 2007. doi: 10.1007/s10708-007-9111-y.
- [20] GraphHopper. GraphHopper routing engine, version 11.0, 2025. URL <https://github.com/graphhopper/graphhopper/releases/tag/11.0>. Released 14 October 2025; Apache License 2.0.
- [21] Mordechai Haklay. How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets. *Environment and Planning B: Planning and Design*, 37(4):682–703, 2010. doi: 10.1068/b35097.
- [22] Benjamin Herfort, Sven Lautenbach, João Porto de Albuquerque, Jennings Anderson, and Alexander Zipf. A spatio-temporal analysis investigating completeness and inequalities of global urban building data in OpenStreetMap. *Nature Communications*, 14:3985, 2023. doi: 10.1038/s41467-023-39698-6.
- [23] Institute for Applied Autonomy. iSee: Paths of least surveillance in Manhattan, 2001. URL <https://www.acmi.net.au/works/109419--isee/>.
- [24] Elizabeth E. Joh. The new surveillance discretion: Automated suspicion, big data, and policing. *Harvard Law & Policy Review*, 10(1):15–42, 2016.
- [25] Mei-Po Kwan. The uncertain geographic context problem. *Annals of the Association of American Geographers*, 102(5):958–968, 2012. doi: 10.1080/00045608.2012.687349.

- [26] Cynthia Lum, Julie Hibdon, Breanne Cave, Christopher S. Koper, and Linda Merola. License plate reader (LPR) police patrols in crime hot spots: An experimental evaluation in two adjacent jurisdictions. *Journal of Experimental Criminology*, 7(4):321–345, 2011. doi: 10.1007/s11292-011-9133-9.
- [27] Cynthia Lum, Christopher S. Koper, James Willis, Stephen Happeny, Heather Vovak, and Jordan Nichols. The rapid diffusion of license plate readers in US law enforcement agencies. *Policing: An International Journal*, 42(3):376–393, 2019. doi: 10.1108/PIJPSM-04-2018-0054.
- [28] Cynthia Lum, Christopher S. Koper, Hyunji Lee, Daniel S. Nagin, and Lawrence Sherman. Measuring the cost-effectiveness of new technologies in policing: The case of automatic license plate readers (ALPR). *Cambridge Journal of Evidence-Based Policing*, 9(1):4, 2025. doi: 10.1007/s41887-025-00099-y.
- [29] Nikita Mazurov. Flock wants the most detailed map of its surveillance cameras taken down. The Intercept, September 2026. URL <https://theintercept.com/2026/09/24/how-many-flock-devices-in-united-states-300000/>. 24 September 2026; accessed 2026-09-25.
- [30] Torin Monahan. Grounding the Flock: Confronting police surveillance of mobilities. *Mobile Media & Communication*, 2026. doi: 10.1177/20501579261453519. Online first.
- [31] New York Civil Liberties Union. Who’s watching? video camera surveillance in New York City and the need for public oversight. NYCLU special report, 2006. URL https://assets.nyclu.org/publications/nyclu_pub_whos_watching.pdf.
- [32] Roland M. Olbricht. Data retrieval for small spatial regions in OpenStreetMap. In Jamal Jokar Arsanjani, Alexander Zipf, Peter Mooney, and Marco Helbich, editors, *OpenStreetMap in GIScience*, Lecture Notes in Geoinformation and Cartography, pages 101–122. Springer, Cham, 2015. doi: 10.1007/978-3-319-14280-7_6.
- [33] OpenStreetMap contributors. OpenStreetMap, 2026. URL <https://www.openstreetmap.org>. Data available under the Open Database License 1.0, <https://www.openstreetmap.org/copyright>.
- [34] OpenStreetMap contributors. OpenStreetMap full-history planet file, 2026. URL <https://planet.openstreetmap.org/planet/full-history/>. Dump of 14 September 2026; Open Database License 1.0.
- [35] OpenStreetMap Wiki contributors. Tag:man_made=surveillance, 2026. URL https://wiki.openstreetmap.org/wiki/Tag:man_made=surveillance. Accessed 2026-09-25.
- [36] OsmAnd BV. OsmAnd: OSM automated navigation directions, 2026. URL <https://github.com/osmandapp/OsmAnd>. Code under GPLv3; artwork and UI/UX under CC-BY-NC-ND 4.0 (repository LICENSE).
- [37] Erich W. Schienke and Institute for Applied Autonomy. On the outside looking out: An interview with the Institute for Applied Autonomy (IAA). *Surveillance & Society*, 1(1):102–119, 2002. doi: 10.24908/ss.v1i1.3396.
- [38] John A. Shjarback and James A. Sarkos. An evaluation of a major expansion in automated license plate reader (ALPR) technology. *Justice Evaluation Journal*, 8(2):225–242, 2025. doi: 10.1080/24751979.2025.2473363.

- [39] Jay Stanley. Fast-growing company Flock is building a new AI-driven mass-surveillance system. American Civil Liberties Union white paper, March 2022. URL <https://www.aclu.org/publications/fast-growing-company-flock-building-new-ai-driven-mass-surveillance-system>.
- [40] U.S. Census Bureau. 2024 Gazetteer files: Census tracts, 2024. URL <https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.2024.html>.
- [41] U.S. Census Bureau. LEHD origin-destination employment statistics (LODES), version 8. Longitudinal Employer-Household Dynamics Program, 2025. URL https://lehd.ces.census.gov/data/.od_main_JT00 files, format version 8.4; accessed 2026-09-23.
- [42] Valhalla contributors. Turn-by-turn parameter “exclude_polygons” doesn’t seem to work, 3.4.0. GitHub issue #4659, 2024. URL <https://github.com/valhalla/valhalla/issues/4659>. Opened 26 March 2024; open as of 2026-09-25.
- [43] Valhalla contributors. Valhalla finding route within excluding polygon. GitHub issue #5069, 2025. URL <https://github.com/valhalla/valhalla/issues/5069>. Opened 23 January 2025; closed 22 December 2025, fixed by #5233.

A Every state

Table 7: Every state and the District of Columbia. Commuters are in-state commuters in the LODS frame; “routed” excludes the unroutable commutes of Section 5.6. Mapped ALPR are nodes inside the state on 2026-09-22, per 100,000 residents from ACS 2022. Exposure and avoidance figures are within-state estimates; the national row is commuter-weighted as in Table 3.

State	Commuters (M)	Commutes routed	Mapped ALPR	per 100k	Mean passed	≥ 1 (%)	Zero when avoiding (%)	Extra min	Over- head (%)
Alabama	1.79	1,159	2,829	56.3	4.76	79.6	82.4	2.83	9.1
Alaska	0.26	608	5	0.7	0.00	0.0	100.0	0.00	0.0
Arizona	2.88	1,150	2,696	37.6	3.47	72.7	93.2	1.10	4.6
Arkansas	1.06	1,146	1,068	35.4	2.84	67.4	92.6	1.09	3.7
California	16.82	1,171	20,932	53.2	7.35	86.7	80.8	2.40	8.8
Colorado	2.54	1,117	2,688	46.6	3.86	79.8	89.2	1.77	7.4
Connecticut	1.41	1,183	1,013	28.1	4.38	72.8	93.2	1.34	6.5
Delaware	0.34	1,104	457	46.0	2.34	68.7	91.7	0.81	4.1
District of Columbia	0.20	1,032	151	22.5	0.93	49.4	92.6	0.00	0.0
Florida	8.80	1,170	9,797	45.3	6.25	86.3	71.3	2.63	9.9
Georgia	4.19	1,182	9,956	92.8	11.78	91.9	54.3	7.78	25.7
Hawaii	0.55	835	85	5.9	0.58	31.5	95.9	0.00	0.0
Idaho	0.69	1,007	463	25.0	2.27	55.6	94.9	0.25	1.2
Illinois	5.29	1,188	8,483	66.5	9.16	90.6	90.5	3.37	15.1
Indiana	2.72	1,176	3,696	54.5	4.72	81.0	92.7	3.16	13.3
Iowa	1.27	1,150	893	28.0	1.63	53.0	96.7	0.00	0.0
Kansas	1.08	1,166	2,018	68.7	4.30	77.6	89.7	1.77	8.9
Kentucky	1.61	1,160	1,915	42.5	3.51	73.6	87.6	1.67	5.7
Louisiana	1.68	1,116	2,088	45.0	4.59	78.7	68.2	3.55	14.1
Maine	0.52	1,066	65	4.8	0.56	26.2	99.0	0.00	0.0
Maryland	2.13	1,183	1,023	16.6	1.86	49.7	90.3	0.00	0.0
Massachusetts	3.12	1,187	1,201	17.2	2.58	65.4	91.5	0.61	2.9
Michigan	3.79	1,190	3,687	36.7	4.61	72.8	94.7	1.70	6.5
Minnesota	2.49	1,163	1,516	26.6	2.26	62.0	95.3	0.39	1.5
Mississippi	0.97	1,125	1,149	38.8	2.79	61.3	89.4	0.67	1.8
Missouri	2.39	1,182	3,453	56.1	5.09	83.0	81.3	2.68	11.0
Montana	0.40	789	87	8.0	0.22	11.8	99.9	0.00	0.0
Nebraska	0.80	1,125	554	28.3	1.29	46.2	98.7	0.00	0.0
Nevada	1.33	1,132	961	30.9	2.51	65.2	95.4	0.46	2.4
New Hampshire	0.52	1,105	67	4.9	0.21	14.8	100.0	0.00	0.0
New Jersey	3.53	1,184	2,410	26.1	3.20	67.5	92.4	0.76	3.1
New Mexico	0.72	994	1,047	49.6	4.19	74.6	83.2	1.41	6.9
New York	8.09	1,194	4,452	22.3	7.42	80.8	62.8	2.42	12.2
North Carolina	4.21	1,179	3,765	36.0	3.46	66.4	88.2	0.94	2.7
North Dakota	0.30	977	390	50.2	2.35	45.9	99.5	0.00	0.0
Ohio	4.84	1,195	6,866	58.3	5.25	84.7	81.9	2.81	10.9
Oklahoma	1.43	1,168	1,493	37.6	2.19	68.4	96.1	0.41	1.1
Oregon	1.68	1,133	541	12.8	0.72	35.4	98.9	0.00	0.0
Pennsylvania	5.16	1,178	3,737	28.8	3.51	60.2	94.9	0.37	1.3
Rhode Island	0.38	1,142	325	29.7	3.06	81.4	83.7	1.46	10.5
South Carolina	1.93	1,159	2,110	41.0	4.59	76.0	83.1	2.12	7.0
South Dakota	0.33	906	163	18.3	0.81	41.2	99.7	0.00	0.0
Tennessee	2.75	1,182	3,619	52.3	3.31	73.9	85.3	1.64	4.3
Texas	12.37	1,176	17,043	58.3	6.89	84.9	81.2	2.26	10.1
Utah	1.49	1,091	1,063	32.4	3.23	81.1	95.7	1.71	8.4
Vermont	0.23	940	30	4.7	0.08	5.4	100.0	0.00	0.0
Virginia	3.30	1,171	3,454	40.0	4.65	73.5	86.1	1.15	4.5
Washington	3.19	1,145	2,301	29.9	2.50	63.9	92.1	0.49	1.6
West Virginia	0.52	1,118	246	13.7	1.14	34.0	93.4	0.00	0.0
Wisconsin	2.53	1,176	2,134	36.3	3.00	71.7	92.9	1.52	6.9
Wyoming	0.20	656	72	12.5	0.51	23.0	92.7	0.00	0.0
United States	132.8	56,131	142,257	27	5.14	75.3	84.2	1.62	6.7